

Assignment-1

(1) Apply **Linear Interpolation method** to estimate the Interpolated value of y when $x = 4$, using the following data points:

$$(x_0, y_0) = (2, 4), (x_1, y_1) = (6, 8).$$

(2) Apply **Lagrange's Interpolation method** using the following data points. Find the Interpolated value of y when $x=40$.

$$\begin{aligned} x &: 30, 35, 45, 55 \\ y &: 148, 96, 68, 34 \end{aligned}$$

(3) Apply **Newton's Forward Interpolation method** using the following data points. Construct the forward difference table and find the Interpolated value of y when $x = 12$.

$$\begin{aligned} x &: 10, 15, 20, 25, 30, 35 \\ y &: 35.3, 32.4, 29.2, 26.1, 23.2, 20.5 \end{aligned}$$

(4) Apply **Newton's Backward Interpolation** method using the following data points. Construct the backward difference table and find the value of y when **Interpolated value** $x=42$.

$$\begin{aligned} x &: 20, 25, 30, 35, 40, 45 \\ y &: 354, 332, 291, 260, 231, 204 \end{aligned}$$

(5) Apply **Trapezoidal Rule** to evaluate the integral of $f(x) = 1 / (1 + x^2)$ from $x = 0$ to $x = 1$, using $n = 4$ sub-intervals. Find value of integral.

(6) Apply Simpson's 1/3rd Rule to evaluate the integral of $f(x) = e^x$ from $x = 0$ to $x = 1$ using $n = 6$ sub-intervals.

(7) Apply Simpson's 3/8 Rule to evaluate the integral of $f(x) = \sqrt{x}$ from $x = 0$ to $x = 4.5$ using $n = 9$ sub-intervals

(8) Use the Euler Method to solve the differential equation

$$dy/dx = x^2 - y$$

Given that $y = 1$ at $x = 0$, find the approximate value of y at $x = 0.1$ using step size $h = 0.05$.

(9) Using the Modified Euler Method, find y at $x = 0.1$ for the differential equation $dy/dx = x^2 + y$, given $y(0) = 1$ and step size $h = 0.05$.